

COMPUTER PROGRAMMING

LAB 4

LAB TITLE

Switch Statement, Break and Continue in C++

1. Objective

- The objective of this lab is to understand how the switch statement works in C++. In this lab, students learn how to select one option from many choices using switch. We also learn how the break statement controls the flow of the program and how the continue statement works inside loops. This lab also helps students create simple menu-driven programs.

2. Theory

- In C++, sometimes a program needs to choose one option from many choices. For this purpose, the switch statement is used. It is often used instead of many if-else statements when comparing one variable with different values.
- Inside a switch statement, different options are written as cases. The program checks the value of the variable and runs the matching case.
- The break statement is used to stop the execution of a case. If break is not used, the program will continue running the next cases.
- The default statement runs when none of the cases match the user input.
- The continue statement is mainly used in loops. It skips the current iteration and moves to the next iteration of the loop.

3. Software

- DEV-C++ IDE
- C++ Compiler (included in DEV-C++)

Program 1: Day Name Using Switch

Algorithm

1. Start the program
2. Declare an integer variable for the day number
3. Ask the user to enter a number from 1 to 7

4. Use a switch statement to check the number
5. Display the corresponding day name
6. End the program

Program 2: Grade Evaluation Using Switch

Algorithm

1. Start the program
2. Declare a character variable for grade
3. Ask the user to enter a grade
4. Use a switch statement to check the grade
5. Display the message according to the grade
6. End the program

Program 3: Unit Conversion Menu Using Switch

Algorithm

1. Start the program
2. Display a menu with different conversion options
3. Take the user's choice
4. Ask the user to enter a value
5. Perform the selected conversion
6. Display the result
7. End the program

Conclusion

- In this lab, we learned how to use the switch statement for making multiple decisions in C++. We also understood the importance of the break statement in controlling the flow of the program. This lab helped us create simple menu-based programs and improved our understanding of conditional structures in C++.